

Definition 28 as a quiver: one drawing decision and what it proves

Shingai Thornton*

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“One could argue that control is always so circular.”

— C. J., margin comment on the Summer 2025 paper, p. 4

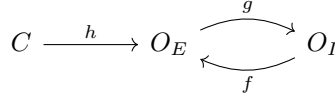
The encoding below takes that sentence literally: it draws control_2 as a cycle. Everything in §2 follows from that one decision, and §4 asks whether the decision is faithful to what you meant. §1 is the only part that is a reading of your text; §2 is machine-checked; §3 lists what §2 does not claim.

1 The encoding

Definition 1 (the quiver Q_J). *Vertices $\{C, O_E, O_I\}$. Generating arrows*

$$h: C \rightarrow O_E, \quad g: O_E \rightarrow O_I, \quad f: O_I \rightarrow O_E.$$

Convention. *Arrows point in the functional direction of J95 Figure 1 (p. 107): an arrow $x \rightarrow y$ means “ x acts on y .”*
Read aloud: C disturbs O_E ; O_E constrains O_I ; O_I feeds back to O_E .



symbol	role in Q_J	J95 (page)
C	vertex	“the global environment” (107); “a source of variation to the system O ” (107)
O_E	vertex	“some efferent component of O , the mechanism whose actions assert compensating variation on O_I ” (107)
O_I	vertex	“the controlled variables” (107); “an internal reflection or representation of any external controlled variable” (107)
h	arrow	“The variation of C affects O through the relation h ” (106)
g	arrow	“the final constraint ₁ of O_E on O_I ” (106)
f	arrow	“whatever relation there might be back from O_I to O_E ” (106)

Definition 2 (the shape). $\mathcal{I}_J := \text{Paths}(Q_J)$, the free category on Q_J : objects are the vertices, morphisms are finite directed paths, composition is concatenation, identities are empty paths. No relations are imposed.¹

Remark 1 (what a quiver forgets). Q_J records only that a named dependency exists. It does not carry: (a) that f, g, h are functions rather than relations (J95 §4.3, p. 114: “usually considered to be functions”); (b) the asymmetry of the loop (§4.3, p. 114: f and g “are not thereby symmetric or complementary”); (c) the hierarchy of C over O and the “large loop gain” (p. 108); (d) the second loop. §3.7.2 (p. 110) says “there is also a kind of loop between C and O_E ” by reciprocal variation, and §3.7.1 (p. 109) rules it one-way at the level of action (“ O_E mirror the actions of C , but not vice versa”). Q_J follows §3.7.1: h is a single arrow. (e) Turchin’s alternative, in which the effector relation g “flows only through the environment” (§3.5, p. 108). Not drawn. Items (d) and (e) are encoding forks, not omissions by oversight; each yields a different Q .

2 The theorem

Write $\ell := f \circ g \in \text{Hom}_{\mathcal{I}_J}(O_E, O_E)$ for the feedback loop, a path of length 2.²

Lemma 1. *For $n \in \mathbb{N}$, ℓ^n has length $2n$; hence $n \mapsto \ell^n$ is injective and $\text{Hom}_{\mathcal{I}_J}(O_E, O_E)$ is infinite.*³

Proof. Length is additive under concatenation, so $|\ell^n| = n|\ell| = 2n$. Distinct n give distinct lengths, hence distinct paths. $\square$

*Drafted with Claude (Fable 5.1) from the Lean sources and from J95 pp. 101–114; the encoding decision, the readings of J95, and the question in §4 are mine. Sources: J95 = C. Joslyn, “Semantic Control Systems,” *World Futures* 45 (1995) 87–123. Lean: `systems-science-foundations, Systems/Category/{ShapeJoslyn, CyclicObstruction, JoslynIncomparability}.lean`, checkout 4288793.

¹`JoslynShape := Paths JoslynPosition, ShapeJoslyn.lean`. `Paths` and `Path.length.comp` are Mathlib’s.

²`joslynLoop, joslynLoop.length : joslynLoop.length = 2 := rfl`.

³`cyclePow.length, cyclePow.injective, infinite.path_of_cycle`, all proved for an arbitrary quiver and an arbitrary positive-length loop.

Theorem 1. *Let $\mathcal{C}$ be any category in which every hom-set is finite. No functor $F: \mathcal{I}_J \rightarrow \mathcal{C}$ is faithful.*⁴

Proof. A faithful functor is injective on each hom-set. By Lemma 1 it would inject the infinite set $\text{Hom}(O_E, O_E)$ into the finite set $\text{Hom}(FO_E, FO_E)$. $\square$

Corollary 1. *Let $\mathcal{I}_K := \text{Paths}(R \rightarrow T)$ be the shape of Klir’s $S = (T, R)$: two objects, one generating arrow. Every hom-set of $\mathcal{I}_K$ is a subsingleton, hence finite. There is no faithful functor $\mathcal{I}_J \rightarrow \mathcal{I}_K$.*⁵

Remark 2 (why this is the boundary and not a curiosity). Every other tradition encoded in the same program (Klir, Bunge, Mobus, Myers, Wymore, Mesarović) has an acyclic Q , hence finite hom-sets, and is placed relative to the others by faithful comparison functors. Q_J is the first cyclic shape in the landscape; the only other is Spivak’s value loop, which instantiates the same lemma. Feedback is where the finite-shape method stops; Theorem 1 makes that folklore a theorem whose sole premise is the cycle.

3 What the theorem does not say

1. **It is presentation-relative.** From the theorem’s own docstring, written before anyone asked you: “both shapes are encodings of primary texts as dependency quivers, and a different defensible encoding of Joslyn 1995 — one that does not make the control loop a cycle at quiver level — could dissolve the obstruction. Not a maximality result.”
2. **It is about Paths, which imposes no relations.** Impose one and it can die. If feedback is taken to be idempotent, $\ell \circ \ell = \ell$, then $\text{Hom}(O_E, O_E)$ collapses to $\{\text{id}, \ell\}$ and Theorem 1 fails. Whether control_2 licenses that identification is a claim about your text, not about category theory.
3. **Re-bracketing does not dissolve it.** Your (31) $\langle O_E, \langle C, O_I \rangle \rangle$ and (32) $\langle O_I, \langle C, O_E \rangle \rangle$ regroup the vertices; neither removes an arrow, so the loop and the theorem survive both. (32) adds a second loop if the “kind of loop” of §3.7.2 is drawn, which only strengthens the obstruction.
4. **Unrolling does dissolve it.** Index the loop by level: vertices $C, O_E^{(k)}, O_I^{(k)}$ for $k \in \mathbb{N}$; arrows $h: C \rightarrow O_E^{(0)}, g_k: O_E^{(k)} \rightarrow O_I^{(k)}, f_k: O_I^{(k)} \rightarrow O_E^{(k+1)}$. This Q is acyclic, every hom-set is finite (at most one path between any two vertices), and faithful functors into $\mathcal{I}_K$ are no longer excluded on cardinality grounds. *Stated, not machine-checked.*
5. **It does not say control_2 is incomparable in principle.** It says the comparison method that places the six acyclic traditions cannot reach a cyclic one. A different method (a quotient, an enrichment, a 2-categorical treatment of the loop) is not ruled out and is not attempted here.

4 The question, which only you can answer

Items 2 through 4 of §3 are one question: *where does the closure of control_2 live?* In the vocabulary of your 2026 abstract: is the semantic closure level-flat, so that each ℓ^n is a genuinely distinct control trajectory, or does it unroll across a metasystem transition, so that the cycle is an artefact of drawing every level on one page? Three outcomes, each a result:

1. **You defend the cycle.** Theorem 1 becomes author-endorsed, the strongest provenance a formalization can carry.
2. **You propose an acyclic encoding** (the unrolling of §3 item 4, or another). The obstruction dissolves for that encoding; presentation-relativity is exhibited rather than asserted.
3. **You rule both encodings defensible.** One text, two shapes, different comparison behaviour: the sharpest exhibit yet for the layer of validation no proof checker sees.

5 One provenance item, separately

Definition 28 is entry 009 of the definition atlas at `math.systems`, transcribed from the page images and hand-checked, with five of your case rulings (thermostat and the decoupled-envelope pair admitted; damped spring, boiling liquid, spinning top refused). One grading is offered for correction: the catalogue records a definition as a *restatement* when its author presents it as another’s, and Definition 25 (control_1) is introduced on p. 101 as Turchin’s 1992 definition, “we offer it again here.” It is therefore graded a restatement of Turchin, and Definition 28 is recorded as the first control definition that is yours alone. If that reading is wrong, the catalogue exists to receive exactly that correction.

⁴`joslyn_no_faithful_functor`, instance of the general `paths_no_faithful_functor_of_cycle`. Axioms: `propext`, `Classical.choice`, `Quot.sound`. No sorry.

⁵`no_faithful_joslyn_to_klir`; the finiteness hypothesis is discharged by an instance built from `klirHomSubsingleton`, not by inspection in prose.